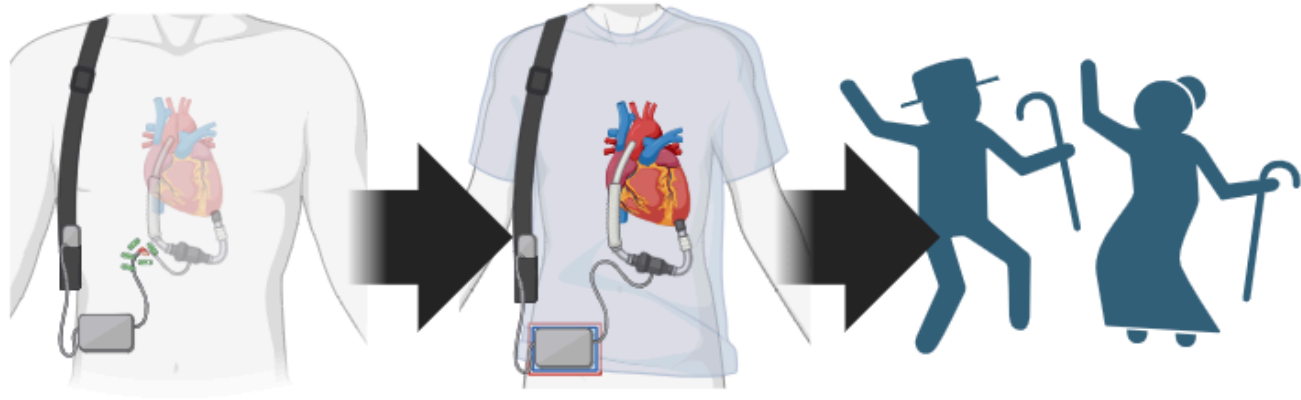


Executive Summary

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The Problem:

High Risk of Sepsis and Driveline Infections:

LVADs, which assist in heart function, require a driveline to power the device. This creates an entry point for infection, leading to complications such as sepsis.

Proposed Innovation:

Polymagnetically-Coupled Inductive Charging System:

Replaces the external driveline with a wireless power system that charges the LVAD through the patient's skin.

Expected Outcome/Improvement:

- Reduction in Driveline infections
- Enhanced Patient Comfort and Quality of Life
- Lower Hospital Readmissions

This project addresses a critical issue in heart failure management, specifically complications associated with current LVAD systems, such as sepsis and driveline infections. This problem is highly significant because it affects many patients and contributes to frequent hospital readmissions and substantial healthcare costs. The proposed innovation of a polymagnetically coupled inductive charging system has the potential to revolutionize LVAD technology by making it fully implantable, thus eliminating driveline infections and improving the quality of life for patients. Incorporating dynamic power delivery and adaptive tuning techniques to manage coil alignment and temperature further demonstrates a commitment to patient comfort and safety. Overall, this project promises to make a meaningful impact on the field of cardiac assist devices.

This project introduces an innovative approach that could fundamentally shift the current LVAD technology landscape. The polymagnetically coupled inductive charging system, called PoLyVAD, represents a novel application of magnet technology not seen in current LVAD systems. The specific use of smart magnets with unique north-south sequences to ensure proper alignment of the charging coils is particularly inventive. This approach addresses a known limitation of existing wireless charging methods, setting our project apart from other emerging fully implantable LVAD systems currently in development. Additionally, incorporating dynamic power delivery and adaptive temperature regulation demonstrates a sophisticated, forward-thinking approach to addressing both technical and patient comfort issues. Overall, our project demonstrates a strong significance and innovative approach to advancing LVAD technology.

Product Comparison			
Solution	Non-Invasive	No Drivelines for Risk of Infection	Targeted Magnetic Technology
poLyVAD	✓	✓	✓
Abbott's Fully Implantable Left Ventricular Assist System (FILVAS)	✓	✓	✗
HeartMate3	✓	✗	✗

PoLyVAD's Technology

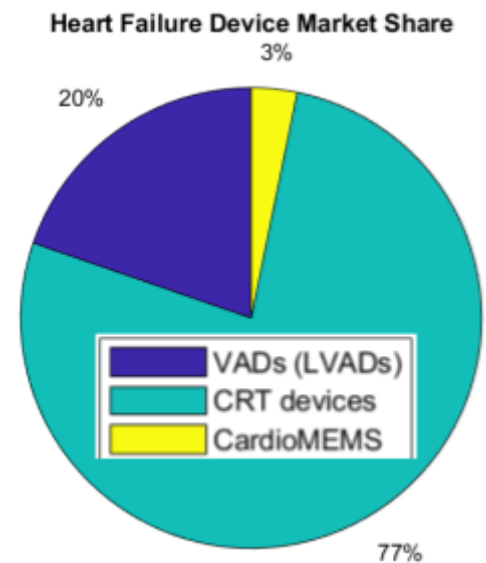
Polymagnetically-Coupled Inductive Charging System: The use of specific north-south magnetic sequencing in smart magnets that ensures the transmitter and receiver coils can only align with each other. This prevents misalignment and avoids accidental attachment to common ferromagnetic objects.

Wireless Inductive Power Transfer System: The specific design of the coils (both transmitter and receiver) used for the wireless power transfer, especially if the shape, size, or materials used enhance efficiency and reduce power loss or heat generation.

Fully Implantable LVAD System: The integration of our wireless charging system with the LVAD's pump, battery, and control system, focuses on how your system operates seamlessly without the need for external cables or penetrative drivelines. This includes the design of the skin interface where the external and internal magnetic components interact, focusing on reducing irritation or complications at the skin surface, and the wearable vest design with an embedded transmitter coil that holds the external controller and transmitter coil, especially how it enhances comfort and ensures stable alignment with the magnetic plate containing a charging coil that is surgically implanted against the inside of the skin.

Market Research

The global market for LVAD systems is experiencing rapid growth. According to BCC Research, the global market for heart failure device technologies was valued at \$5.7 billion in 2020 and is projected to reach \$8.8 billion by 2025, with a compound annual growth rate (CAGR) of 9.1% during this period [1]. This growth is driven by LVADs being an essential solution for patients with advanced heart failure or those ineligible for heart transplants. In the same report, VADs, which include LVADs, account for \$1.1 billion or 20.2% of the emerging heart failure device market, followed by CRT devices at 76.6% of the market at \$4.3 billion, and CardioMEMS, valued at \$179 million, as shown in the figure. The clinical benefits of making fully implantable LVADs, which can reduce infection rates and improve patient quality of life, make them an increasingly attractive option for healthcare providers and investors.



FDA Regulation

The LVAD proposed in this project is classified by the FDA as a “Ventricular (Assist) Bypass,” under the product code DSQ. As such, it will follow the FDA's premarket approval application (PMA) pathway for Class III medical devices [2], overseen by the Office of Cardiovascular Devices (OHT2) at the FDA. The technical content of the PMA application requires both non-clinical laboratory investigations and clinical studies.

We look to existing FDA-approved LVAD devices as a basis for the standards of premarket approval. Most durable continuous-flow LVADs are first approved for use as a bridge-to-transplant (BTT) device, for patients awaiting permanent heart transplants. Devices are approved for longer-term therapeutic use after further clinical testing. The FDA approved the Abbott HeartMate 3 LVAD device for both use cases in 2017 [3]. It has largely replaced its predecessor, the Thoratec HeartMate 2. Its main competitor, Medtronic's HVAD system has been discontinued for future implants. This leaves the HeartMate 3 as the dominant player in the LVAD device market [3], as well as the best option for this project to use as a model for its premarket approval application.

Clinical Study & Statistical Analysis

Our project's testing pathway will begin with bench testing of the wireless induction charging system to assess heat dissipation, power transfer efficiency, durability, and the potential for electromagnetic interference with the rest of the LVAD system. Following device validation, we will conduct animal studies to evaluate the LVAD's integration with the circulatory system, biocompatibility, and long-term stability. These animal studies will be performed on pigs and will focus on the risk of inflammation, thrombosis, and complications related to wireless charging. After achieving positive outcomes in our animal testing, we will proceed to human clinical

trials. Smaller Phase I trials will demonstrate device safety, and later Phase II and Phase III trials will evaluate long-term efficacy and patient outcomes compared to existing devices. The HeartMate 3 took around six years to get approved for clinical use by the FDA, and we aim to mimic that timeline with our tests.

During the PMA approval process of the HeartMate 3, Abbott sponsored a clinical study, MOMENTUM 3 that compared the HeartMate 3 to the earlier HeartMate 2. This randomized trial, involving 1026 patients, indicated superior outcomes for the HeartMate 3. Alongside preclinical testing, this clinical data was instrumental in securing FDA approval [3]. With a similarly sized and structured equivalence study, our project will demonstrate that it meets the safety and efficacy benchmarks required for PMA approval.

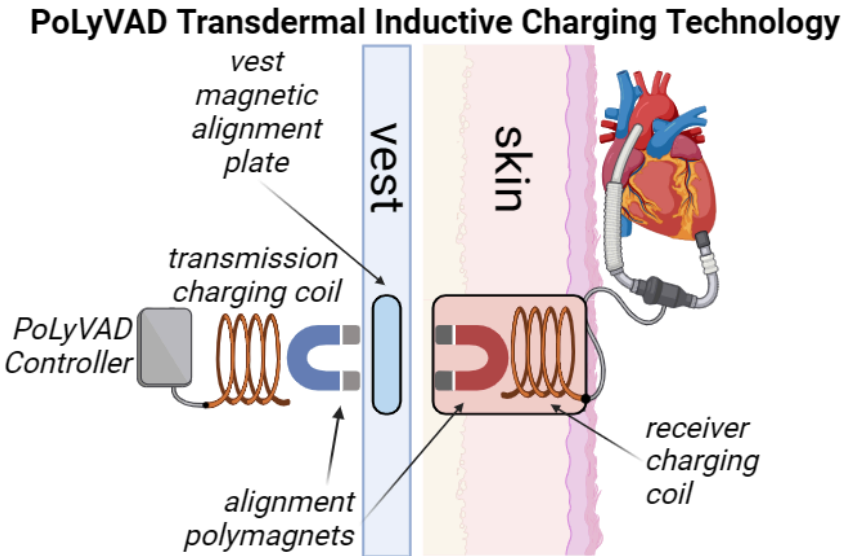
Intellectual Property

Since our proposed innovation is a functional improvement, we will apply for a utility-type patent and this gives us a 20-year term from the effective filing date. As our core inventive concept has already been established and is ready to present, we will take action to file a provisional/non-provisional application to the U.S. patent office.

For our proposed innovation, we will be focusing on patenting the specific design and functionality of the smart magnetic coupling system and any unique aspects related to how it integrates with the LVAD. What will be patented is PoLyVAD’s technology previously listed above:

Fully implantable LVAD system: We will be patenting the setup in what makes PoLyVAD unique such as the wearable vest with alignment plate design and the polymagnetically-coupled inductive charging system. This includes the material composition or structural design/distinctive pattern of these polymagnets, particularly if they have unique properties that improve alignment or durability over time.

Wireless inductive power transfer system: We will also patent the temperature regulation system that dynamically adjusts power delivery to prevent overheating. This includes safety and stability features like software algorithms or hardware configurations that manage energy transmission to keep battery temperature at or below body temperature.



Funding Strategy

Non-dilutive: For this project, we plan to apply for many grants and secure multiple funding sources. This innovation would be of interest to the NIH, specifically the National Heart, Lung, and Blood Institute (NHLBI). We will apply for an R01 grant, due to the immense amount of preliminary data about LVAD. Along with our own group’s preliminary research, an R01 could be successful and give us the most money to progress this project. Additionally, since several companies are currently attempting to advance LVAD, we will apply for Medtronic or Abbott grants and collaborations to improve on ongoing work at these companies. This may be a more feasible option for us since we would need expertise from LVAD experts to bring this innovation to life.

Dilutive: By analyzing data from PitchBook and identifying similar companies and their investors, we can draw inspiration from the investment portfolios of these promising LVAD medical startups. Companies like ReliantHeart, which is developing a remotely monitored implantable cardiological support device, and Evaheart, another LVAD developer, have already secured funding from prominent venture capitalists. Vivo Capital and Innova Capital are key investors for these companies and both have an emphasis on investing in the healthcare sector. By strategically targeting venture capitalists with a proven track record in funding similar medical devices such as Vivo Capital and Innova Capital we can position our product for successful investment and growth.

References

- [1] "Global Markets for Emerging Medical Device Technologies." *BCC Research*, 2020, <https://www.bccresearch.com/market-research/healthcare/emerging-medical-device-markets-report.html>.
- [2] "Product Classification." *Accessdata.Fda.Gov*, Food and Drug Administration, www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfPCD/classification.cfm?ID=DSQ.
- [3] Stehlik, Josef, and James K. Kirklin. "The long and winding road to an effective left ventricular assist device: The demise of Medtronic's Hvac." *American Heart Association*, vol. 144, no. 7, 17 Aug. 2021, pp. 509–511, <https://doi.org/10.1161/circulationaha.121.056027>.